

Chemical Reactions and Equations

Class 10 Science - Chapter 1 (NCERT)

1. Introduction to Chemical Reactions

A chemical reaction is a fundamental process where one or more substances, known as reactants, are transformed into entirely different substances called products [1, 2]. This transformation occurs because chemical bonds between atoms in the reactants are broken and new bonds are formed to create the products [2].

In any chemical reaction, the substances that undergo the change are the reactants, while the new substances formed are the products [1]. Common examples include the rusting of iron and the burning of magnesium in the air [3].

- Reactants: Substances that undergo chemical change [1].
- Products: New substances formed after the reaction [1].
- Example: $2Mg + O_2 \rightarrow 2MgO$ (Burning of Magnesium) [3].

2. Observations: How to Identify a Chemical Reaction

We can determine if a chemical reaction has taken place by observing specific changes in the substances involved [2]. These observable changes are clear indicators that a chemical transformation is occurring [4].

Common observations include a change in state or color, such as copper turning black when heated [4]. Other indicators are the evolution of a gas, like hydrogen bubbles forming when zinc reacts with acid, or a change in temperature, which indicates heat is being released or absorbed [4, 5]. Additionally, the formation of a precipitate (an insoluble solid) is a strong sign of a reaction [5].

- Change in color or state (e.g., Copper turns black) [4].
- Evolution of gas (e.g., H_2 gas bubbles) [4].
- Change in temperature (Exothermic or Endothermic) [5].
- Formation of a precipitate (insoluble solid) [5].

3. Chemical Equations and Balancing

A chemical equation is a symbolic representation of a chemical reaction using formulas and symbols [1]. The reactants are written on the left side and products on the right, separated by an arrow ($\rightarrow$) pointing toward the products to show the direction of the reaction [3, 6].

Balancing chemical equations is crucial to satisfy the Law of Conservation of Mass, which states that matter cannot be created or destroyed [6]. Therefore, the number of atoms of each element must remain the same on both the reactant and product sides [6, 7]. To balance an equation, we adjust the coefficients (the numbers placed before formulas) without changing the chemical formulas themselves [6].

- Unbalanced: $H_2 + O_2 \rightarrow H_2O$ (Oxygen is unbalanced) [8].
- Balanced: $2H_2 + O_2 \rightarrow 2H_2O$ (All atoms are equal) [8].
- Coefficients must be in the simplest whole number ratio [6].

4. Types of Chemical Reactions

Chemical reactions are classified into different types based on how reactants transform into products. The five main types studied in Class 10 are Combination, Decomposition, Displacement, Double Displacement, and Redox reactions [9].

- Combination: Two or more reactants form a single product ($2H_2 + O_2 \rightarrow 2H_2O$) [9].
- Decomposition: A single compound breaks down into simpler substances ($2H_2O_2 \rightarrow 2H_2O + O_2$) [9].
- Displacement: One element replaces another in a compound ($Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$) [9].
- Double Displacement: Exchange of ions between two compounds ($Na_2SO_4 + BaCl_2 \rightarrow BaSO_4 + 2NaCl$) [9].

5. Oxidation and Reduction (Redox Reactions)

A reaction where one substance is oxidized and another is reduced is called a Redox (Oxidation-Reduction) reaction [9]. Oxidation is defined as the gain of oxygen or the loss of electrons,

while reduction is the loss of oxygen or the gain of electrons [9].

For example, in the reaction $ZnO + C \rightarrow Zn + CO$, Carbon (C) gains oxygen to become Carbon Monoxide (CO), meaning it is oxidized [9]. Conversely, Zinc Oxide (ZnO) loses oxygen to become Zinc (Zn), meaning it is reduced [9]. Identifying which substance is oxidized and which is reduced is a key skill in this chapter [10].

- Oxidation: Gain of Oxygen or Loss of Electrons [9].
- Reduction: Loss of Oxygen or Gain of Electrons [9].
- Redox Example: $CuO + H_2 \rightarrow Cu + H_2O$ (H_2 is oxidized; CuO is reduced) [10].

6. Effects of Oxidation in Everyday Life

Oxidation reactions have significant effects in our daily lives, primarily seen as corrosion and rancidity. Corrosion is the gradual destruction of metals due to reactions with moisture, acids, or air [5]. Common examples include the rusting of iron (reddish-brown coating), the blackening of silver, and the green coating on copper [5].

Rancidity occurs when fats and oils in food oxidize, resulting in an unpleasant smell and taste [11]. To prevent this, foods containing fats are often kept in airtight containers or flushed with inert gases like nitrogen (e.g., in chip bags) to slow down the oxidation process [11].

- Corrosion: Rusting of iron ($4Fe + 3O_2 \rightarrow 2Fe_2O_3$) [3, 5].
- Rancidity: Oxidation of fats causing bad taste/smell [11].
- Prevention: Use antioxidants or nitrogen flushing for foods [11].

7. Key Questions and Reasoning

There are several conceptual questions frequently asked regarding chemical reactions. For instance, magnesium ribbon must be cleaned before burning to remove the protective oxide layer on its surface, allowing it to burn efficiently [10].

Another common experiment involves dipping an iron nail into a copper sulfate solution. The blue solution turns green because iron is more reactive than copper and displaces it,

forming iron sulfate ($FeSO_4$) and depositing reddish-brown copper on the nail [11]. This is a classic example of a displacement reaction [9, 11].

- Why clean Magnesium ribbon? To remove the obstructing oxide layer [10].
- Iron nail in $CuSO_4$: Solution turns from blue to green due to formation of $FeSO_4$ [11].
- White Precipitate: Reaction of $BaCl_2$ and Na_2SO_4 forms insoluble $BaSO_4$ [12].

